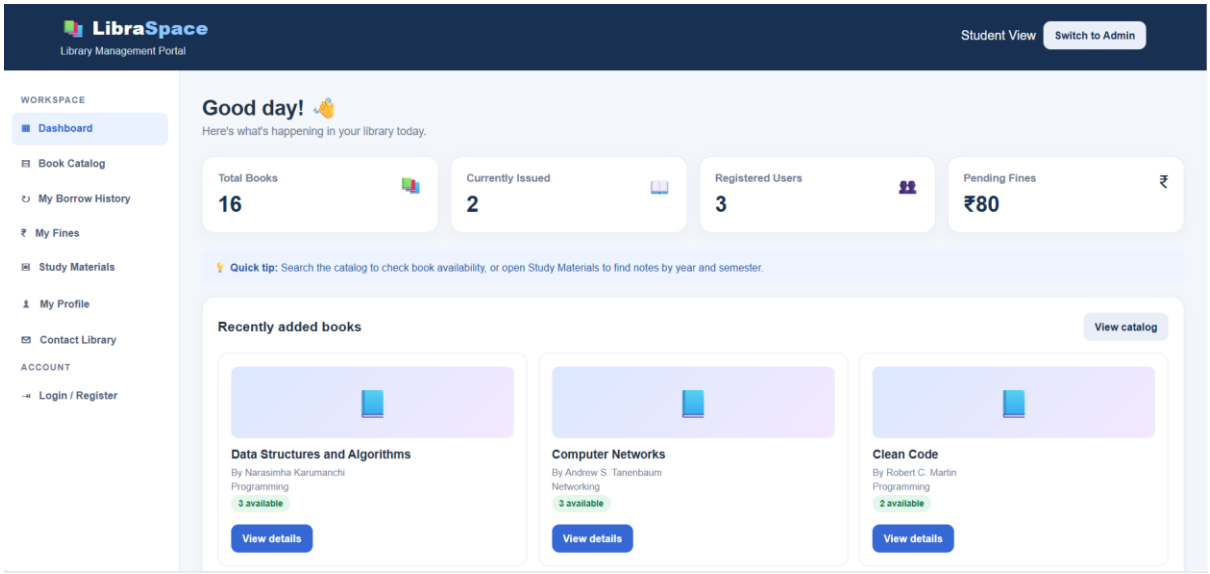
1. A local library needs a website to streamline operations and enhance user experience. The site will include a catalog for searching and viewing books, user accounts for borrowing and managing books, and an admin dashboard for managing inventory and user data. Essential features include a search function, book details page, user login system, and contact form, The system helps both students and library managers to keep constant track of all the books available in the library. It allows both the admin and the student to search for the desired book. It becomes necessary for colleges to keep a continuous check on the books issued and returned and even calculate fines.




```
from pathlib import Path

p = Path("/mnt/data/library_management.html")

html = r'''<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>LibraSpace | Library Management</title>

<style>

:root{--navy:#183153;--blue:#3568d4;--pale:#f3f6fb;--ink:#243247;--muted:#718096;--white:#fff;--green:#198754;--red:#d9485f;--line:#e5eaf2}
```

*{box-sizing:border-box}body{margin:0;font-family:Arial,Helvetica,sans-serif;background:var(--pale);color:var(--ink)}

header{background:var(--navy);color:white;padding:17px 5%;display:flex;align-items:center;justify-content:space-between;gap:15px;flex-wrap:wrap}

.brand{font-size:23px;font-weight:800;letter-spacing:.3px}.brand span{color:#8fc6ff}

header p{margin:5px 0 0;color:#d8e4f4;font-size:13px}.header-actions{display:flex;gap:9px;align-items:center}

button,.btn{border:0;border-radius:9px;padding:10px 14px;background:var(--blue);color:white;cursor:pointer;font-weight:700;font-size:13px}

button:hover,.btn:hover{filter:brightness(.94)}button.secondary{background:#e9eef8;color:var(--navy)}button.danger{background:var(--red)}button.success{background:var(--green)}

.layout{display:grid;grid-template-columns:225px 1fr;min-height:calc(100vh - 78px)}

aside{background:#fff;padding:22px 14px;border-right:1px solid var(--line)}.nav-label{font-size:11px;color:var(--muted);font-weight:bold;margin:9px 12px 10px;text-transform:uppercase;letter-spacing:1px}

.nav-btn{display:block;width:100%;text-align:left;background:transparent;color:#506078;padding:12px;border-radius:9px;margin:3px 0;font-weight:600}.nav-btn.active,.nav-btn:hover{background:#eaf1ff;color:var(--blue)}

main{padding:28px;min-width:0}.page-title{margin:0 0 6px;font-size:26px}.sub{color:var(--muted);margin:0 0 22px;font-size:14px}

.stats{display:grid;grid-template-columns:repeat(4,minmax(0,1fr));gap:15px;margin-bottom:22px}.stat,.panel{background:#fff;border:1px solid var(--line);border-radius:15px;padding:19px;box-shadow:0 4px 15px #223b5b08}.stat small{color:var(--muted);font-weight:600}.stat strong{display:block;font-size:27px;margin-top:9px;color:var(--navy)}.stat .icon{float:right;font-size:22px}

.panel{margin-bottom:18px}.panel-head{display:flex;align-items:center;justify-content:space-between;gap:12px;flex-wrap:wrap;margin-bottom:15px}.panel h2{font-size:17px;margin:0}.toolbar{display:flex;gap:9px;flex-wrap:wrap}.toolbar input,.toolbar select{width:auto;min-width:170px}

input,select,textarea{width:100%;padding:11px 12px;border:1px solid #d9e1ed;border-radius:9px;font:inherit;color:var(--ink);background:#fff}textarea{min-height:95px;resize:vertical}label{display:block;font-size:12px;font-weight:700;color:#53627a;margin:12px 0 6px}.form-grid{display:grid;grid-template-columns:repeat(2,minmax(0,1fr));gap:0 13px}.form-actions{display:flex;gap:9px;margin-top:15px;flex-wrap:wrap}

table{border-collapse:collapse;width:100%;font-size:13px}th,td{padding:13px 10px;text-align:left;border-bottom:1px solid var(--line);vertical-align:middle}th{color:#687890;background:#f8faff;font-size:11px;text-transform:uppercase;letter-spacing:.4px}td{line-height:1.45}.table-wrap{overflow-x:auto}.pill{display:inline-block;border-radius:20px;padding:5px 9px;font-size:11px;font-

weight:bold;background:#e4f7ec;color:#137546}.pill.warn{background:#fff1d6;color:#966000}.pill.red{background:#ffe6e9;color:#b82e43}

.book-grid{display:grid;grid-template-columns:repeat(3,minmax(0,1fr));gap:14px}.book-card{border:1px solid var(--line);border-radius:12px;padding:16px;background:#fff}.book-cover{height:90px;border-radius:9px;background:linear-gradient(135deg,#dbe8ff,#f3e8ff);display:grid;place-items:center;font-size:35px;margin-bottom:12px}.book-card h3{font-size:15px;margin:0 0 6px}.book-card p{font-size:12px;color:var(--muted);margin:4px 0}.book-card button{margin-top:11px}

.notice{background:#edf4ff;color:#2856a5;border-radius:10px;padding:12px 14px;font-size:13px;margin-bottom:17px}.hidden{display:none!important}.empty{padding:24px;text-align:center;color:var(--muted)}footer{text-align:center;color:var(--muted);padding:20px;font-size:12px}

@media(max-width:950px){.stats{grid-template-columns:repeat(2,1fr)}.book-grid{grid-template-columns:repeat(2,1fr)}.layout{grid-template-columns:190px 1fr}main{padding:20px}}

@media(max-width:650px){.layout{display:block}aside{display:flex;overflow-x:auto;padding:8px;gap:5px;border-right:0;border-bottom:1px solid var(--line)}.nav-label{display:none}.nav-btn{white-space:nowrap;width:auto;margin:0;padding:10px}.stats{grid-template-columns:repeat(2,1fr)}.book-grid{grid-template-columns:1fr}main{padding:16px}.form-grid{grid-template-columns:1fr}.page-title{font-size:22px}header{padding:14px 16px}}

</style>

</head>

<body>

<header>

📚 LibraSpace Library Management Portal

Student View<button class="secondary" onclick="toggleRole()">Switch to Admin</button>

</header>

<aside>

Workspace

<button class="nav-btn active" data-page="dashboard" onclick="showPage('dashboard',this)">▦   Dashboard</button>

<button class="nav-btn" data-page="books" onclick="showPage('books',this)">▤   Book Catalog</button>

<button class="nav-btn" data-page="borrow" onclick="showPage('borrow',this)">↻   My Borrow History</button>

<button class="nav-btn" data-page="fines" onclick="showPage('fines',this)">₹   My Fines</button>

<button class="nav-btn" data-page="materials" onclick="showPage('materials',this)">▣   Study Materials</button>

<button class="nav-btn" data-page="profile" onclick="showPage('profile',this)">♗   My Profile</button>

<button class="nav-btn" data-page="contact" onclick="showPage('contact',this)">✉   Contact Library</button>

Admin Tools

<button class="nav-btn admin-only hidden" data-page="manage" onclick="showPage('manage',this)">⚙   Manage Books</button>

<button class="nav-btn admin-only hidden" data-page="issue" onclick="showPage('issue',this)">⇄   Issue / Return</button>

<button class="nav-btn admin-only hidden" data-page="manageMaterials" onclick="showPage('manageMaterials',this)">＋   Manage Materials</button>

<button class="nav-btn admin-only hidden" data-page="users" onclick="showPage('users',this)">♔   User Data</button>

Account

<button class="nav-btn" onclick="openAuth()">⇥   Login / Register</button>

</aside>

<main>

<section id="page-dashboard" class="page">

<h1 class="page-title">Good day! 👋</h1> Here's what's happening in your library today.

📚<small>Total Books</small>6

📖<small>Currently Issued</small>2

👥<small>Registered Users</small>3

₹<small>Pending Fines</small>₹0

💡 Quick tip: Search the catalog to check book availability, or open Study Materials to find notes by year and semester.

<h2>Recently added books</h2><button class="secondary" onclick="showPage('books')">View catalog</button>

<h2>Quick actions</h2> <button onclick="showPage('books')">Search books</button><button class="secondary" onclick="showPage('borrow')">Borrow history</button><button class="secondary" onclick="showPage('materials')">Find study materials</button>

</section>

<section id="page-books" class="page hidden">

<h1 class="page-title">Book Catalog</h1> Search books by title, author, or category.

<h2>Find a book</h2> <input id="bookSearch" placeholder="Search title, author..." oninput="renderBooks()"><select id="bookFilter" onchange="renderBooks()"><option value="all">All availability</option><option value="available">Available</option><option value="issued">Issued</option></select>

</section>

<section id="page-borrow" class="page hidden">

<h1 class="page-title">My Borrow History</h1> Track issued books, due dates, and return status.

<table><thead><tr><th>Book</th><th>Issue Date</th><th>Expected Return</th><th>Status</th></tr></thead><tbody id="borrowRows"></tbody></table>

</section>

<section id="page-fines" class="page hidden">

<h1 class="page-title">My Fines</h1> Review overdue charges associated with your account.

<h2>Fine summary</h2> No pending fines. <table><thead><tr><th>Book</th><th>Days overdue</th><th>Fine</th><th>Status</th></tr></thead><tbody id="fineRows"></tbody></table>

</section>

<section id="page-materials" class="page hidden">

<h1 class="page-title">Study Materials</h1> Find and download notes, lab manuals, and learning resources.

<h2>Browse materials</h2> <select id="matYear" onchange="renderMaterials()"><option value="all">All years</option><option>1</option><option>2</option><option>3</option><option>4</option></select><select id="matSem" onchange="renderMaterials()"><option value="all">All semesters</option><option>1</option><option>2</option><option>3</option><option>4</option><option>5</option><option>6</option><option>7</option><option>8</option></select><input id="matSearch" placeholder="Search material..." oninput="renderMaterials()"> <table><thead><tr><th>Material</th><th>Year</th><th>Semester</th><th>Type</th><th>Action</th></tr></thead><tbody id="materialRows"></tbody></table>

</section>

<section id="page-profile" class="page hidden">

<h1 class="page-title">My Profile</h1> Manage your library account details.

<h2>Personal information</h2> <label>Full name</label><input id="profileName" value="Student User"> <label>Email address</label><input id="profileEmail" type="email" value="student@example.com"> <label>Student ID</label><input id="profileId" value="STU2026001"> <label>Department / Year</label><input id="profileDept" value="CSE / Year 2"> <button onclick="saveProfile()">Save changes</button>

</section>

<section id="page-contact" class="page hidden">

<h1 class="page-title">Contact the Library</h1> Send a question to the librarian.

<form onsubmit="sendContact(event)"><label>Your name</label><input required placeholder="Enter your name"><label>Email address</label><input required type="email" placeholder="you@example.com"><label>Subject</label><input required placeholder="How can we help?"><label>Message</label><textarea required placeholder="Write your message..."></textarea> <button type="submit">Send message</button> </form>

</section>

<section id="page-manage" class="page hidden admin-page">

<h1 class="page-title">Manage Books</h1> Add new books or update and remove catalog records.

<h2 id="bookFormTitle">Add a book</h2><input id="editBookId" type="hidden"> <label>Book title</label><input id="newTitle" placeholder="e.g. Introduction to Java"> <label>Author</label><input id="newAuthor" placeholder="Author name"> <label>Category</label><input id="newCategory" placeholder="e.g. Programming"> <label>Copies available</label><input id="newCopies" type="number" min="0" value="1"> <div

class="form-actions"><button onclick="saveBook()">Save book</button><button class="secondary" onclick="clearBookForm()">Clear</button>

<h2>Inventory</h2><input style="max-width:260px" id="adminBookSearch" placeholder="Search inventory..." oninput="renderAdminBooks()"> <table><thead><tr><th>Title</th><th>Author</th><th>Category</th><th>Available</th><th>Actions</th></tr></thead><tbody id="adminBookRows"></tbody></table>

</section>

<section id="page-issue" class="page hidden admin-page">

<h1 class="page-title">Issue & Return</h1> Record book loans and mark books as returned.

<h2>Issue a book</h2> <label>Student name</label><input id="issueStudent" placeholder="Student name"> <label>Book</label><select id="issueBook"></select> <label>Issue date</label><input id="issueDate" type="date"> <label>Expected return date</label><input id="dueDate" type="date"> <button onclick="issueBook()">Issue book</button>

<h2>Active loans</h2> <table><thead><tr><th>Student</th><th>Book</th><th>Issue date</th><th>Due date</th><th>Action</th></tr></thead><tbody id="adminLoanRows"></tbody></table>

</section>

<section id="page-manageMaterials" class="page hidden admin-page">

<h1 class="page-title">Manage Study Materials</h1> Add a resource with year and semester details. Use a public file URL to enable downloads.

<h2>Add material</h2> <label>Material name</label><input id="newMatTitle" placeholder="Data Structures Notes"> <label>File type</label><select id="newMatType"><option>PDF</option><option>DOCX</option><option>PPT</option><option>Link</option></select> <label>Year</label><select id="newMatYear"><option>1</option><option>2</option><option>3</option><option>4</option></select> <label>Semester</label><select id="newMatSem"><option>1</option><option>2</option><option>3</option><option>4</option><option>5</option><option>6</option><option>7</option><option>8</option></select> <label>Download URL (optional)</label><input id="newMatUrl" type="url" placeholder="https://example.com/notes.pdf"> <button onclick="addMaterial()">Add material</button>

<h2>Uploaded materials</h2> <table><thead><tr><th>Material</th><th>Year</th><th>Semester</th><th>Action</th></tr></thead><tbody id="adminMaterialRows"></tbody></table>

</section>

<section id="page-users" class="page hidden admin-page">

<h1 class="page-title">User Data</h1> Sample registered users for the admin view. <table><thead><tr><th>Name</th><th>Email</th><th>Student ID</th><th>Role</th></tr></thead><tbody id="userRows"></tbody></table>

</section>

</main>

<footer>LibraSpace Library Management Portal · Front-end demo for academic use</footer>

<button class="secondary" style="position:absolute;right:14px;top:14px" onclick="closeAuth()">X</button><h2>Welcome to LibraSpace</h2> Sign in or create a student account. <button id="loginTab" onclick="setAuthTab('login')">Login</button><button id="registerTab" class="secondary" onclick="setAuthTab('register')">Register</button> <form onsubmit="authSubmit(event)"><label>Full name (registration only)</label><input id="authName" placeholder="Your name"><label>Email</label><input id="authEmail" type="email" required placeholder="you@example.com"><label>Password</label><input id="authPassword" type="password" required minlength="4" placeholder="At least 4 characters"> <button type="submit" id="authSubmitBtn">Login</button> </form> Demo only: authentication is not secure and data is stored in this browser session.

<script>

let isAdmin=false, authMode='login', currentUser={name:'Student User',email:'student@example.com',id:'STU2026001',dept:'CSE / Year 2'};

let books=[

{id:1,title:'Introduction to Java',author:'Herbert Schildt',category:'Programming',copies:3,available:2},

{id:2,title:'Database System Concepts',author:'Abraham Silberschatz',category:'Database',copies:2,available:1},

{id:3,title:'Operating System Concepts',author:'Abraham Silberschatz',category:'Computer Science',copies:2,available:2},

{id:4,title:'Clean Code',author:'Robert C. Martin',category:'Programming',copies:2,available:2},

{id:5,title:'Computer Networks',author:'Andrew S. Tanenbaum',category:'Networking',copies:3,available:3},

```
{id:6,title:'Data Structures and Algorithms',author:'Narasimha Karumanchi',category:'Programming',copies:4,available:3}
];
let loans=[
{id:1,student:'Student User',bookId:1,issue:'2026-10-15',due:'2026-10-29',status:'Issued'},
{id:2,student:'Student User',bookId:2,issue:'2026-10-01',due:'2026-10-12',status:'Issued'}
];
let materials=[
{id:1,title:'Programming in Java - Notes',year:'2',sem:'3',type:'PDF',url:''},
{id:2,title:'Database Management Systems',year:'2',sem:'4',type:'PDF',url:''},
{id:3,title:'Engineering Mathematics',year:'1',sem:'1',type:'PDF',url:''},
{id:4,title:'Computer Networks Lab Manual',year:'3',sem:'5',type:'PDF',url:''}
];
let users=[
{name:'Student User',email:'student@example.com',id:'STU2026001',role:'Student'},
{name:'Ananya Kumar',email:'ananya@example.com',id:'STU2026002',role:'Student'},
{name:'Ravi Kumar',email:'ravi@example.com',id:'STU2026003',role:'Student'}
];
let nextBookId=7,nextLoanId=3,nextMaterialId=5;
function safe(s){return String(s??'').replace(/[&<>"']/g,c=>({'&':'&amp;','<':'&lt;','>':'&gt;','"':'&quot;',"'":'&#39;'}[c]));}
function bookName(id){return books.find(b=>b.id===id)?.title||'Deleted book';}
function showPage(page,btn){
 document.querySelectorAll('.page').forEach(p=>p.classList.add('hidden'));
 document.getElementById('page-'+page)?.classList.remove('hidden');
 document.querySelectorAll('.nav-btn[data-page]').forEach(b=>b.classList.toggle('active',b.dataset.page===page));
 if(btn)btn.classList.add('active');refresh();
}
function toggleRole(){isAdmin=!isAdmin;document.getElementById('roleLabel').textContent=isAdmin?'Librarian Admin View':'Student View';document.querySelectorAll('.admin-only').forEach(e=>e.classList.toggle('hidden',!isAdmin));if(isAdmin){showPage('manage');}else{s
```

howPage('dashboard');}document.querySelector('header .header-actions button').textContent=isAdmin?'Switch to Student':'Switch to Admin';}

function refresh(){renderStats();renderBooks();renderBorrow();renderFines();renderMaterials();renderAdminBooks();renderLoans();renderAdminMaterials();renderUsers();renderRecent();populateIssueBooks();}

function renderStats(){document.getElementById('statBooks').textContent=books.reduce((n,b)=>n+b.copies,0);document.getElementById('statIssued').textContent=loans.filter(l=>l.status==='Issued').length;document.getElementById('statUsers').textContent=users.length;document.getElementById('statFines').textContent='₹'+loans.filter(l=>l.status==='Issued').reduce((n,l)=>n+fineFor(l),0);}

function renderRecent(){document.getElementById('recentBooks').innerHTML=books.slice(-3).reverse().map(bookCard).join('')||' No books yet. ';}

function bookCard(b){return `<div class="book-card"><div class="book-cover">📘</div><h3>${safe(b.title)}</h3><p>By ${safe(b.author)}</p><p>${safe(b.category)}</p><p>${b.available} available</p><button onclick="alert('Book: ${safe(b.title)}\\nAuthor: ${safe(b.author)}\\nCategory: ${safe(b.category)}\\nCopies available: ${b.available}')">View details</button></div>`;}

function renderBooks(){let q=(document.getElementById('bookSearch')?.value||'').toLowerCase(),f=document.getElementById('bookFilter')?.value||'all';let list=books.filter(b=>(b.title+' '+b.author+' '+b.category).toLowerCase().includes(q)&&(f==='all'||(f==='available'?b.available>0:b.available===0)));document.getElementById('bookCatalog').innerHTML=list.map(bookCard).join('')||' No matching books found. ';}

function renderBorrow(){let list=loans.filter(l=>l.student.toLowerCase()===currentUser.name.toLowerCase());document.getElementById('borrowRows').innerHTML=list.map(l=>`<tr><td>${safe(bookName(l.bookId))}</td><td>${safe(l.issue)}</td><td>${safe(l.due)}</td><td><span class="pill ${l.status==='Returned'?'':'warn'}">${safe(l.status)}</span></td></tr>`).join('')||'<tr><td colspan="4" class="empty">No borrowing history found.</td></tr>';}

function fineFor(l){if(l.status==='Returned')return 0;let days=Math.max(0,Math.floor((new Date().setHours(0,0,0,0)-new Date(l.due+'T00:00:00').getTime())/86400000));return days*5;}

function renderFines(){let list=loans.filter(l=>l.student.toLowerCase()===currentUser.name.toLowerCase()),total=list.reduce((n,l)=>n+fineFor(l),0);document.getElementById('fineSummary').textContent=total?`Your current estimated fine is ₹${total} (₹5 per overdue day). Please contact the librarian to settle it.`:'No pending fines. You are all caught up!';document.getElementById('fineRows').innerHTML=list.map(l=>{let days=Math.floor(fineFor(l)/5),fine=fineFor(l);return `<tr><td>${safe(bookName(l.bookId))}</td><td>${days}</td><td>₹${fine}</td><td><span

```
class="pill ${fine?'red':''}">${fine?'Pending':'Clear'}</span></td></tr>`}).join('')||'<tr><td colspan="4" class="empty">No fine records.</td></tr>';}

function renderMaterials(){let y=document.getElementById('matYear')?.value||'all',s=document.getElementById('matSem')?.value||'all',q=(document.getElementById('matSearch')?.value||'').toLowerCase();let list=materials.filter(m=>(y==='all'||m.year===y)&&(s==='all'||m.sem===s)&&m.title.toLowerCase().includes(q));document.getElementById('materialRows').innerHTML=list.map(m=>`<tr><td>${safe(m.title)}</td><td>${safe(m.year)}</td><td>${safe(m.sem)}</td><td>${safe(m.type)}</td><td>${m.url?`<a href="${safe(m.url)}" target="_blank" rel="noopener">Download</a>`:'<span class="pill warn">Demo file</span>'}</td></tr>`).join('')||'<tr><td colspan="5" class="empty">No materials found for these filters.</td></tr>';}

function saveBook(){let title=document.getElementById('newTitle').value.trim(),author=document.getElementById('newAuthor').value.trim(),category=document.getElementById('newCategory').value.trim(),copies=Math.max(0,Number(document.getElementById('newCopies').value)||0),id=Number(document.getElementById('editBookId').value);if(!title||!author||!category){alert('Please enter title, author, and category.');return;}if(id){let b=books.find(b=>b.id===id);if(b){let issued=b.copies-b.available;b.title=title;b.author=author;b.category=category;b.copies=Math.max(copies,issued);b.available=b.copies-issued;}}else{books.push({id:nextBookId++,title,author,category,copies,available:copies});}clearBookForm();refresh();alert('Book saved successfully.');}

function clearBookForm(){['editBookId','newTitle','newAuthor','newCategory'].forEach(id=>document.getElementById(id).value='');document.getElementById('newCopies').value=1;document.getElementById('bookFormTitle').textContent='Add a book';}

function renderAdminBooks(){let q=(document.getElementById('adminBookSearch')?.value||'').toLowerCase();let list=books.filter(b=>(b.title+' '+b.author+' '+b.category).toLowerCase().includes(q));document.getElementById('adminBookRows').innerHTML=list.map(b=>`<tr><td>${safe(b.title)}</td><td>${safe(b.author)}</td><td>${safe(b.category)}</td><td>${b.available}/${b.copies}</td><td><button class="secondary" onclick="editBook(${b.id})">Edit</button> <button class="danger" onclick="deleteBook(${b.id})">Delete</button></td></tr>`).join('')||'<tr><td colspan="5" class="empty">No books found.</td></tr>';}

function editBook(id){let b=books.find(b=>b.id===id);if(!b)return;document.getElementById('editBookId').value=id;document.getElementById('newTitle').value=b.title;document.getElementById('newAuthor').value=b.author;document.getElementById('newCategory').value=b.category;document.getElementById('newCopies').value=b.copies;document.getElementById('bookFormTitle').textContent='Update book';window.scrollTo({top:0,behavior:'smooth'});}

function deleteBook(id){if(loans.some(l=>l.bookId===id&&l.status==='Issued')){alert('Cannot delete a book with an active loan. Mark it returned first.');return;}if(confirm('Delete this book from the catalog?')){books=books.filter(b=>b.id!==id);refresh();}}
```

```
function populateIssueBooks(){let s=document.getElementById('issueBook');if(!s)return;let old=s.value;s.innerHTML=books.filter(b=>b.available>0).map(b=>` <option value="${b.id}">${safe(b.title)} (${b.available} available)</option>` ).join('');if(old)s.value=old;}

function issueBook(){let student=document.getElementById('issueStudent').value.trim(),id=Number(document.getElementById('issueBook').value),issue=document.getElementById('issueDate').value,due=document.getElementById('dueDate').value,b=books.find(b=>b.id===id);if(!student||!b||!issue||!due){alert('Enter student, book, issue date and due date.');return;}if(due<issue){alert('Expected return date cannot be before issue date.');return;}if(b.available<1){alert('No copies available.');return;}b.available--;loans.push({id:nextLoanId++,student,bookId:id,issue,due,status:'Issued'});document.getElementById('issueStudent').value='';refresh();alert('Book issued successfully.');}

function renderLoans(){let list=loans.filter(l=>l.status==='Issued');document.getElementById('adminLoanRows').innerHTML=list.map(l=>` <tr><td>${safe(l.student)}</td><td>${safe(bookName(l.bookId))}</td><td>${safe(l.issue)}</td><td>${safe(l.due)}</td><td><button class="success" onclick="returnBook(${l.id})">Mark returned</button></td></tr>` ).join('')||'<tr><td colspan="5" class="empty">No active loans.</td></tr>';}

function returnBook(id){let l=loans.find(l=>l.id===id&&l.status==='Issued');if(!l)return;let fine=fineFor(l);l.status='Returned';let b=books.find(b=>b.id===l.bookId);if(b)b.available=Math.min(b.copies,b.available+1);refresh();alert('Book returned.'+(fine?` Estimated overdue fine: ₹${fine} (₹5/day).` :' No overdue fine.'));}

function addMaterial(){let title=document.getElementById('newMatTitle').value.trim(),type=document.getElementById('newMatType').value,year=document.getElementById('newMatYear').value,sem=document.getElementById('newMatSem').value,url=document.getElementById('newMatUrl').value.trim();if(!title){alert('Enter a material name.');return;}materials.push({id:nextMaterialId++,title,type,year,sem,url});document.getElementById('newMatTitle').value='';document.getElementById('newMatUrl').value='';refresh();alert('Study material added.');}

function renderAdminMaterials(){document.getElementById('adminMaterialRows').innerHTML=materials.map(m=>` <tr><td>${safe(m.title)}</td><td>${safe(m.year)}</td><td>${safe(m.sem)}</td><td><button class="danger" onclick="deleteMaterial(${m.id})">Delete</button></td></tr>` ).join('')||'<tr><td colspan="4" class="empty">No materials added.</td></tr>';}

function deleteMaterial(id){if(confirm('Delete this material?')){materials=materials.filter(m=>m.id!==id);refresh();}}

function renderUsers(){document.getElementById('userRows').innerHTML=users.map(u=>` <tr><td>${safe(u.name)}</td><td>${safe(u.email)}</td><td>${safe(u.id)}</td><td>${safe(u.role)}</td></tr>` ).join('');}
```

```
function
saveProfile(){currentUser={...currentUser,name:document.getElementById('profileName').value
.trim()||'Student
User',email:document.getElementById('profileEmail').value.trim(),id:document.getElementById
('profileId').value.trim(),dept:document.getElementById('profileDept').value.trim()};alert('Profile
updated for this demo.');refresh();}

function
openAuth(){document.getElementById('authModal').classList.remove('hidden');document.getEl
ementById('authModal').style.display='flex';}

function closeAuth(){document.getElementById('authModal').style.display='none';}

function
setAuthTab(mode){authMode=mode;document.getElementById('authName').parentElement.q
uerySelector('label').style.display=mode==='register'?'block':'none';document.getElementById(
'authName').style.display=mode==='register'?'block':'none';document.getElementById('loginTa
b').className=mode==='login'?'':'secondary';document.getElementById('registerTab').classNa
me=mode==='register'?'':'secondary';document.getElementById('authSubmitBtn').textContent
=mode==='login'?'Login':'Create account';}

function authSubmit(e){e.preventDefault();let
email=document.getElementById('authEmail').value.trim(),name=document.getElementById('a
uthName').value.trim();if(authMode==='register'){if(!name){alert('Enter your name to
register.');return;}if(users.some(u=>u.email.toLowerCase()===email.toLowerCase())){alert('An
account with this email already exists.');return;}let
u={name,email,id:'STU'+String(2026000+users.length+1),role:'Student'};users.push(u);currentU
ser={name,email,id:u.id,dept:'Not set'};alert('Demo account created.');}else{let
u=users.find(u=>u.email.toLowerCase()===email.toLowerCase());if(!u){alert('Demo login: use
student@example.com, or register a new
account.');return;}currentUser={name:u.name,email:u.email,id:u.id,dept:'CSE / Year
2'};alert('Welcome back,
'+u.name+'!');}document.getElementById('profileName').value=currentUser.name;document.ge
tElementById('profileEmail').value=currentUser.email;document.getElementById('profileId').val
ue=currentUser.id;closeAuth();refresh();}

function sendContact(e){e.preventDefault();alert('Thank you! Your message is ready in this
demo. A backend is needed to send it to the librarian.');e.target.reset();}

let today=new Date(),dueDefault=new
Date(today);dueDefault.setDate(today.getDate()+14);document.getElementById('issueDate').va
lue=today.toISOString().slice(0,10);document.getElementById('dueDate').value=dueDefault.toI
SOString().slice(0,10);

setAuthTab('login');refresh();

</script>

</body>

</html>'''

p.write_text(html, encoding="utf-8")
```

```
print(p, p.exists(), p.stat().st_size)
```